

SPECIFICATION FOR 0-80 KV, AC HV TEST SET

1.0 INTRODUCTION:

The utility of this will be for the testing of applied voltage test on CT, PT, Transformers, Motors, Insulators etc. at various stages of manufacturing and/or erection at different applied voltages.

2.0 TECHNICAL DATA

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| 2.1 Input supply | : 0-230 V $\pm$ 10% volts, 1 Phase, 50 $\pm$ 5% Hz, AC supply. |
| 2.2 Output voltage | : Continuously variable 0 to 80 KV AC. |
| 2.3 Capacity | : 100mA in 0-80 KV range |
| 2.4 Duty Cycle | : 5 min On, 10 min Off. |
| 2.5 Accuracy of TR. | : $\pm$ 3 % or better |
| 2.6 Accuracy of Ammeter, Voltmeter | : $\pm$ 1 % or better |
| 2.7 Percentage Impedance | : not more than 8% in any case. |
| 2.8 Output voltage variation | : By means of motor driven arrangement. |
- Over load Tripping adjustment through site selection: At 10mA, 20mA, 50mA and 100mA AC.
 - Over Voltage Indication & Interlock Protection provided.
 - Normal Over Load Tripping set at 50 mA

3.0 FEATURES OF 0-80 KV, AC HV TEST KIT

The equipment must be in two main units to be provided with wheels for easy movement as described below :

3.1 Control and Metering unit :

It should consist of main input fuses, MCB, contactor fuses, supply indication lamps, push buttons, timers, automatic overload tripping circuits 'EMERGENCY OFF' pushbutton., meters etc. housed in a robust cabinet with adequate ventilation and should have good aesthetic appearance. Digital meters of reputed make for indication as well as for measurement are preferred over analog. The cables/ wiring to be terminated on suitable terminal blocks with crimped lugs and ferrule nos. Gland plate should be provided at the bottom or rear side for cable entry. The whole control circuitry will be housed within a sheet metal cabinet treated properly and powder coated. Continuously variable autotransformer (Rated to operate on 0-230 V, 50 Hz) of suitable capacity (Current capacity shall not be less than 40Amp.) to increase/decrease the output test voltage up to the required voltage level. INCREASE/DECREASE circuit to control motor operated Voltage Regulator. The continuously variable Auto Transformer shall be natural air cooled. When high voltage circuit is switched ON the regulator will rotate to increase the output voltage when INCREASE / DECREASE switch is kept in 'INCREASE' position and 'INCREASE' push is pressed ON or decrease the output voltage when the 'INCREASE/ DECREASE' switch is kept in 'DECREASE' position.

3.1.1 Protection& Interlock:

- Enclosure interlocking will be provided i.e. high voltage circuit cannot be energised unless the door of the enclosure is closed. However, this interlock can be by-passed by shorting corresponding terminals. Zero start interlocking will be provided to ensure that the HV circuit cannot be energized unless the regulator is initially kept or brought back to zero position.
- Automatic tripping mechanism for protecting the HV transformer against over loading. The tripping mechanism can be adjusted at values as mentioned above. After tripping, High Voltage will be switched OFF.
- In case the output voltage crosses the maximum rated value a lamp & a buzzer will indicate overvoltage and the increase mode circuit will not function further, even if 'PRESS TO INCREASE' push is pressed.

The instruments for indications and measurements include following :

Voltmeter	: Digital Voltmeter generally connected to the tertiary winding of HV Transformer for measurement and scaled in KV. Resolution: 0.1KV.
Ammeter	: Digital milli ammeter along with 'PRESS TO READ mA' push to read the leakage current at HT side. Resolution: 0.1mA.
KV meter	: For output voltage measurement
Timer	: A 'TIMER' (0-999secs.) with start switch will be provided to count down the duration of applied HV. After completion of the test time the regulator will automatically decrease to zero position. 'RESET' feature of the 'TIMER' will be provided.

3.2 High voltage transformer :

- Oil cooled step up transformer specially designed to cater HVbreakdown applications.
- The HV winding of the Transformer is of graded insulation; one end being connected to the earth potential through a CT operated Ammeter & tripping device, while the other end remains in floating condition as HT Terminal. The details of tripping arrangement is also to be clearly defined.
- The transformer will be designed to withstand frequent intermittent spark over or short circuit conditions under which such testing transformers are designed to operate.
- HT terminal will be provided with anti-corona metallic ring of adequate diameter to facilitate uniform charge distribution.

3.2.1 **Core:** The core of the transformer will be of high grade CRGO steel of M4, M3 or M2 H grade material only.

3.2.2 **Coils :** The coils will be made of 99.99 % pure electrolytic copper conductor with suitable insulation between turns and layers. The primary and secondary coils will be of round shapes so as to withstand mechanical stress during short circuit and during normal operation.

- 3.2.3 **Wiring** : The wiring will be carried out with suitable cross section multi strand copper conductor. The cable should be terminated with ferrule and crimped lugs on both the ends. The bunch of wire be neatly laid out and secured to the body cabinet with button tape.
- 3.2.4 HV point to be brought out with suitable porcelain insulator and should have isolated LV point brought out on the top of body to assist tan delta measurement with existing equipment.
- 3.2.5 The HV and LV Bushing shall be of porcelain and of reputed make and the HV Bushing shall be of adequate voltage rating to withstand 80 KV. The LV side bushing shall be of 1.1 KV class.

4.0 Qualifying Criterion:

Only OEM of the Item can apply.

The manufacturer must have designed, manufactured, tested and supplied at least Three(3) sets of similar or high end 0-80 KV AC High Voltage Test Set to WBSEDCL / other Power Utilities / other Govt. Departments in earlier occasions within last three financial years. Documents in support of supply of the tendered items to WBSEDCL / other Power Utilities / other Govt. Departments in earlier occasions within last three financial years as mentioned below:

- a) Purchase Orders, Inspection Offer letter, Despatch Instructions, Signed Challan etc. for completing supply of the item against a particular contract.
- b) Successful Type Test Report as per clause no. 6.0 of the specification carried out at third party NABL accredited Laboratory in India or equivalent international laboratories. The Type Test should be carried out on similar Type and Model which is going to be offered.
- c) Performance Certificate from other Power Utilities / other Govt. Departments where the Test Kit was supplied earlier within last Three Years.

If the bidder can submit the Credential of the offered set from Power Utility or Government Departments supplied through Channel partners, those certificates will also be acceptable.

In both cases offered set means Sets with same model nos. as those of offered ones, if available.

5.0 Degree of Protection: IP 53

6.0 Type Test:

The High Voltage P.T. shall be Type Tested from third party NABL accredited Laboratory or from equivalent laboratory of International repute for Degree of Protection IP 53. Those tests are to be conducted within last 05(Five) year.

7.0 **CABLES**

8.1 **Primary cable:**

Primary cables should be 3 core and Minimum 20 Meters long suitable for connection to $230 \pm 10\%$ volts, 50 $\pm$ 5% Hz AC supply. 1100 V grade cable to be used.

9.0 Max. Dimension of unit (L x B x H) to be furnished with the offer.

10.0 Net weight with GA drawing to be furnished with the offer.

11.0 PAINTING:

HV test equipment shall be powder coated outside with epoxy paint light gray RAL-7035 and inside with white epoxy paint and all bright steel components shall be coated with rust preventive paint before dispatch.

12.0 TEST AND INSPECTION

- Supplier to submit his QA plan along with offer.
- Testing shall be done as per relevant standards. Inspection will be done at supplier's works by WBSEDCL representative. Internal testing and QC reports along with one copy instruction manual shall be furnished for WBSEDCL's approval before giving call for inspection.
- Supplier should have all the facility to conduct the following basic tests:-

1. No Load Current of the Auto Transformer.
2. Insulation test of H.V. Transformer.
3. No Load and Load Test on Continuously Variable Auto Transformer.
4. Turns ratio (%Error) measurement of the P.T.

The supplier should submit with offer, the testing scheme together with Line Diagram for all the test required on the above H.V. Tester.

13.0 INSTRUCTION MANUAL:

The instructional manual shall consists of following sections. The instruction manual shall be made on good quality paper (at least 90 gsm) and shall be made in bound volumes (6 copies)suitable for long term usage in shop/site.

- Introduction.
- Photograph of all items of equipment
- Description of equipment, GA drawings, Schematic diagram/ Circuit diagrams.
- Test certificate.
- Detailed procedure to operate the equipment (written in easy language for understanding of operators)
- Dos and Dents, FAQ(Frequently asked Questions) and answers
- Trouble shooting flow chart.
- Service centre contact details like address, email address, phone nos, cell nos. etc.

14.0 GUARANTEE:

- a) All the equipments supplied against this specification shall be guaranteed for a defect liability period of 60 months from the date of receive at DTD store duly passed acceptance testing at DTD lab after delivery.

- b) Any engineering error, omission, wrong provision, equipment failure etc. the equipment shall be attended by the bidder at free of cost during defect liability period.
- c) For any defect in the High Voltage Test Set during defect liability period repairing has to be completed within fifteen days. Otherwise substitute set of same type has to be provided within fifteen days from the date of reporting. It will be handed over back to supplier on repair & receipt of original set.
- d) If any defect observed during defect liability period, it is bidder's responsibility to collect kit from site/ consignee to attend the same.
- e) On each failure of test set, supplier shall submit detailed report of failure analysis to Distribution Testing Department, WBSEDCL.

- The equipment shall be guaranteed for reliable and trouble free operation for 60 months from the date of commissioning & to replace defective parts/equipment/ materials free of cost within guarantee period.

15.0 DOCUMENTATION:

- 1 set complete with drawings/leaflets/catalogue and technical information giving full description, operation, dimensions, weight etc. along with offer.

16.0 COMPLETENESS OF OFFER:

Supplier to confirm compliance of specifications first instance itself. Any deviation shall be clearly brought out in the offer, without this the offer will not be considered and will be rejected.

Guaranteed Technical particulars

Sl. No.	Item to be replied	To be filled up by the Bidder
1	Name of the Manufacturer	
2.	Office Address	
3.	Manufacturing Unit Address	
4.	Whether the bidder is the OEM of the tendered items (Yes/No)	
5.	Whether the bidder has comprehensive service centre along with competent Engineers, if yes, mention the address.	
6.	Model Names of the High voltage P.T. offered by them (Mentioning Item wise Model No., if any)	
7.	What is the voltage ratio of High Voltage P.T.	
8.	Whether the P.T. has the tripping arrangement at selectable leakage current values 10mA, 20 mA, 50 mA & 100 mA.	
9.	What is the Accuracy Class of the P.T.	
10.	What is the make , resolution & Accuracy class of the KV meter?	
11.	What is the make resolution & Accuracy Class of the milli Amp meter	
12.	What is the make resolution & Accuracy Class of the Timer	
13.	What is the make of Bushing and it's voltage class	
14.	What are the Type tests conducted on H.V. P.T.	
15.	What is the make of C.T. and it's type & Accuracy class	

16.	Whether the tripping arrangement of the P.T. has clearly been defined.	
17.	No. of Terminals on LV side	
18.	Weight and overall dimension of the H.V P.T.	
19.	Whether the P.T is oil cooled	
20.	Whether the continuously variable Auto Transformer is natural Air cooled?	
21.	Whether the P.T. is guaranteed for a period of 05(Five) years from the date of it's commissioning.	
22.	i) Whether submitted satisfactory performance of the equipment from any power utility/Govt. department for a period of 3 years as per specification.	
	ii) Whether the bidder has all the Testing facility available at their works.	

Signature of Company representative with seal